Abstract

During natural tactile exploration, interactions between the skin and object surfaces generate complex vibrotactile signals that contain both spectral and temporal information, contributing to texture perception (Manfredi et al., 2014; Le et al., 2024). Although previous research has demonstrated that humans can discriminate complex vibrations and individual stimulus features, relatively little is known about how these features collectively organize perceptual representations. The present study used Multidimensional Scaling (MDS) to investigate the perceptual organization of complex vibrotactile stimuli across three pilot experiments. Stimulus parameters were selected based on previous psychophysical findings. Pilot 1 examined the number of Tones, the inter-frequency interval (IFI), and Amplitude Modulation (AM) Depth. Based on evidence that waveform shape is itself a discriminable tactile cue, Number of Tones was replaced with Waveform Shape in Pilot 2 (Bensmaia & Hollins, 2000). Pilot 3 introduced Attack Time while retaining IFI and AM Depth, following recent findings demonstrating reliable tactile discrimination of temporal acoustic features (Chauvette et al., 2025). Participants completed similarity judgment tasks, and the resulting similarity matrices were analyzed using MDS, along with qualitative verbal descriptions. Across all three pilots, Inter-Frequency Interval (IFI) and Amplitude Modulation (AM) Depth consistently produced stronger perceptual organization than Number of Tones, Waveform Shape, or Attack Time, while participants primarily described stimuli in terms of holistic tactile qualities such as roughness, intensity, continuity, and vibration characteristics. These findings provide an initial characterization of the perceptual space of complex vibrotactile stimuli and establish a refined stimulus set for future studies on vibrotactile categorization and learning.

1. Introduction

Natural tactile interactions generate complex vibrations containing both spectral and temporal information that support the perception of texture and material properties (Manfredi et al., 2014). Unlike simple

sinusoidal stimuli, real-world tactile signals consist of multiple interacting frequency components that are processed simultaneously by the somatosensory system (Le et al., 2024).

Previous studies have identified several perceptually relevant stimulus dimensions. Spectral properties such as the number of Frequency Components and the inter-frequency interval (IFI) influence tactile discrimination (Le et al., 2024), while Waveform Shape provides an additional cue even when frequency content is similar (Bensmaia & Hollins, 2000). More recently, temporal features including Amplitude Modulation (AM), Depth, and Attack Time have also been shown to be reliably perceived through touch (Chauvette et al., 2025).

Although these studies demonstrate that individual stimulus dimensions can be discriminated, they provide limited insight into how multiple stimulus dimensions jointly organize perceptual similarity. Multidimensional Scaling (MDS) addresses this by reconstructing perceptual spaces directly from similarity judgments, allowing underlying perceptual dimensions to emerge without predefined categories (Picard et al., 2003; Gaißert et al., 2010).

The present study employed MDS to examine the perceptual organization of complex vibrotactile stimuli across three iterative pilot experiments. The stimulus dimensions were progressively refined based on previous literature and observations from earlier pilots. Verbal descriptions were collected to aid the interpretation of the recovered perceptual spaces. The objective was to identify stimulus dimensions that produce stable perceptual organization and establish a refined stimulus set for future studies of vibrotactile categorization.

2. Methods

2.1 Participants

Three pilot experiments were conducted with different participant groups. Pilot 1 included five participants, Pilot 2 included six participants, and Pilot 3 included five participants. All participants completed the similarity judgment task independently for the set of vibrotactile stimuli presented in the respective pilot.

2.2 Apparatus

The experiment was conducted using a Hapcoil Plus Long V2 vibrotactile actuator driven by an Actronika 2020 controller and a DW-CT14+ amplifier. Stimuli were generated using custom Python scripts and presented through a custom experimental interface. Participants placed the pad of their left index finger on the actuator throughout the experiment. To minimize auditory cues produced by the actuator, participants wore headphones during stimulus presentation.

2.3 Stimuli

Three pilot experiments were conducted during the project. Each pilot consisted of 27 complex vibrotactile stimuli generated using a full factorial combination of three physical stimulus parameters. Across the three pilots, one stimulus dimension was modified based on observations from the preceding experiment, while the remaining experimental parameters were kept unchanged.

Pilot 1

The first pilot manipulated Tone Count, Inter-Frequency Interval (IFI), and Amplitude Modulation (AM) Depth.

Variable	Levels
Tone Count	2, 3, 4
IFI	50, 100, 150 Hz
AM Depth	0.3, 0.6, 0.9

Pilot 2

Based on the findings of Pilot 1, **Tone Count** was replaced with **Waveform Shape**, while IFI and AM Depth were retained.

Variable	Levels
Waveform	Sine, Square, Sawtooth
IFI	50, 100, 150 Hz
AM Depth	0.3, 0.6, 0.9

Pilot 3

The third pilot replaced **Waveform Shape** with **Attack Time**, introducing a temporal envelope manipulation while retaining IFI and AM Depth.

Variable	Levels
Attack Time	100, 250, 400 ms
IFI	50, 100, 150 Hz
AM Depth	0.3, 0.6, 0.9

For all three pilots, the remaining stimulus parameters were held constant. Stimuli had a duration of 1 s, a base frequency of 80 Hz, an amplitude modulation frequency of 10 Hz, and were generated at a sampling rate of 44.1 kHz. RMS normalization was applied to maintain approximately constant stimulus intensity across all conditions.

The base frequency of 80 Hz was selected to minimize potential confounding arising from differential activation of rapidly adapting (RA) and Pacinian (PC) mechanoreceptive channels. Keeping the base frequency within this range helped maintain relatively consistent mechanoreceptor involvement across stimuli while allowing systematic manipulation of the experimental stimulus dimensions.

2.4 Experimental Procedure

Participants completed a pairwise similarity judgment task. During each trial, two vibrotactile stimuli were presented sequentially, and participants indicated whether the two vibrations felt **similar** or **dissimilar**. They were instructed to base their responses on their overall perceptual impression rather than any specific physical property. Before the experiment began, participants completed several practice trials to familiarize themselves with the task.

Each trial followed the sequence shown below.

Event	Duration
Fixation	300 ms
Stimulus A	1 s
Inter-Stimulus Interval	150 ms
Stimulus B	1 s
Response Window	2 s
Inter-Trial Interval	150 ms

Each pilot consisted of pairwise comparisons between the 27 stimuli. Participant responses were recorded together with reaction times for subsequent analysis.

Following the similarity judgment task in Pilots 2 and 3, participants completed an additional verbal description task. Each stimulus was presented individually in a randomized order, and participants were asked to describe the perceived sensation using one to three words. These qualitative descriptions were later compared with the recovered perceptual spaces.

2.5 Data Analysis

For each participant, similarity judgments were converted into a 27 × 27 similarity matrix, with higher values indicating greater perceived similarity between stimulus pairs. Individual matrices were averaged to obtain a group similarity matrix for each pilot.

The group similarity matrices were converted into dissimilarity matrices and analyzed using non-metric Multidimensional Scaling (MDS) to recover the underlying perceptual space. Both individual and group

MDS solutions were examined. To aid interpretation, the recovered perceptual spaces were coloured according to the manipulated physical stimulus dimensions.

For Pilots 2 and 3, the verbal descriptions were visualized on both the individual and group MDS plots to qualitatively examine whether similar verbal descriptors clustered within the recovered perceptual space.

3. Results

3.1 Pilot 1: Tone Count, Inter-Frequency Interval, and AM Depth

3.1.1 Stress Analysis

To determine the appropriate dimensionality of the recovered perceptual space, non-metric multidimensional scaling (MDS) solutions were computed for one to four dimensions. The stress values decreased substantially from one to two dimensions, with only modest improvements upon adding further dimensions (Figure 1). Consequently, a two-dimensional solution was selected as a suitable compromise between goodness-of-fit and interpretability.

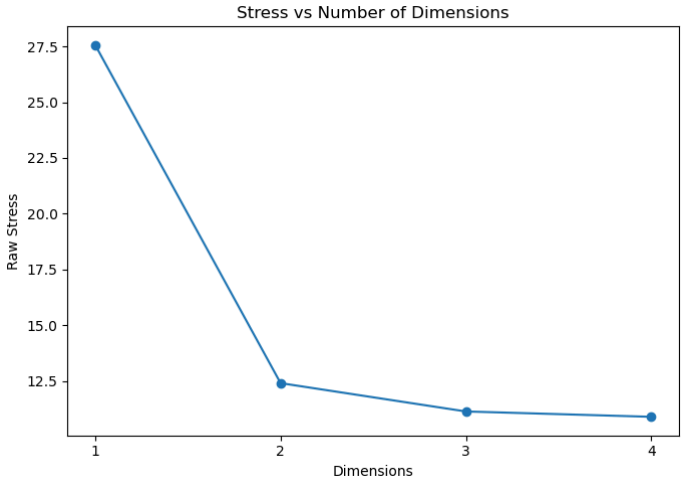


Figure 1. Stress values for one- to four-dimensional MDS solutions for Pilot 1

3.1.2 Group Perceptual Space

The two-dimensional MDS solution recovered a stable perceptual representation of the 27 complex vibrotactile stimuli (Figure 2). Rather than being randomly distributed, the stimuli occupied distinct regions of perceptual space, indicating that participants organized them according to consistent perceptual relationships. At this stage, the recovered dimensions represent latent perceptual dimensions and do not directly correspond to any specific physical stimulus parameter.

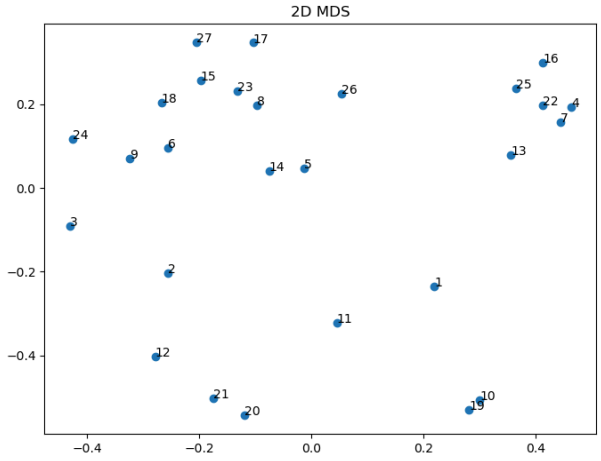


Figure 2. Two-dimensional group MDS configuration for Pilot 1

3.1.3 Relationship Between Physical Stimulus Parameters and Perceptual Space

To facilitate interpretation of the recovered perceptual space, the MDS configuration was colored by each manipulated stimulus parameter (Figure 3).

When colored by Amplitude Modulation (AM) Depth, a clear separation of the three AM levels was observed. Stimuli with similar AM depths tended to occupy neighboring regions of the perceptual space, indicating that AM depth strongly influenced perceived similarity.

Coloring the same configuration by Inter-Frequency Interval (IFI) also revealed an organized gradient, though greater overlap was observed between neighboring IFI levels. This suggests that IFI contributed to perceptual organization but less distinctly than AM depth.

In contrast, coloring the MDS solution according to Tone Count showed comparatively weaker clustering. While some local grouping of stimuli with the same tone count was visible, substantial overlap across the three tone-count conditions indicated that Tone Count exerted a weaker influence on participants' similarity judgments.

Overall, these visualizations suggest that temporal modulation (AM depth) produced the strongest perceptual organization, followed by IFI, whereas Tone Count contributed comparatively less to the recovered perceptual space.

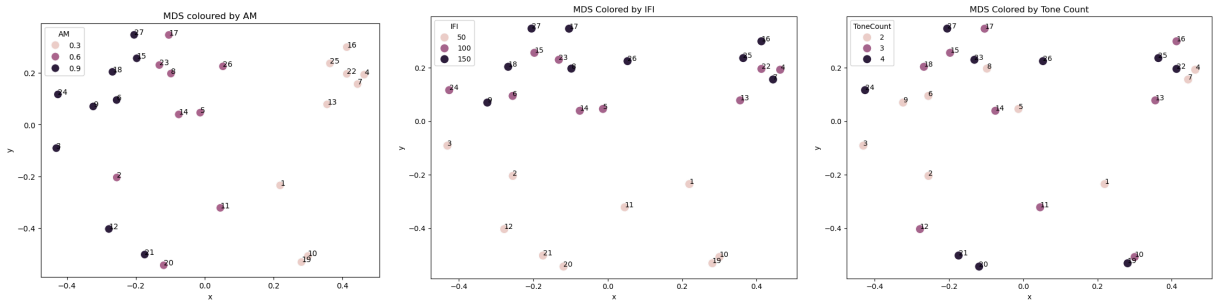
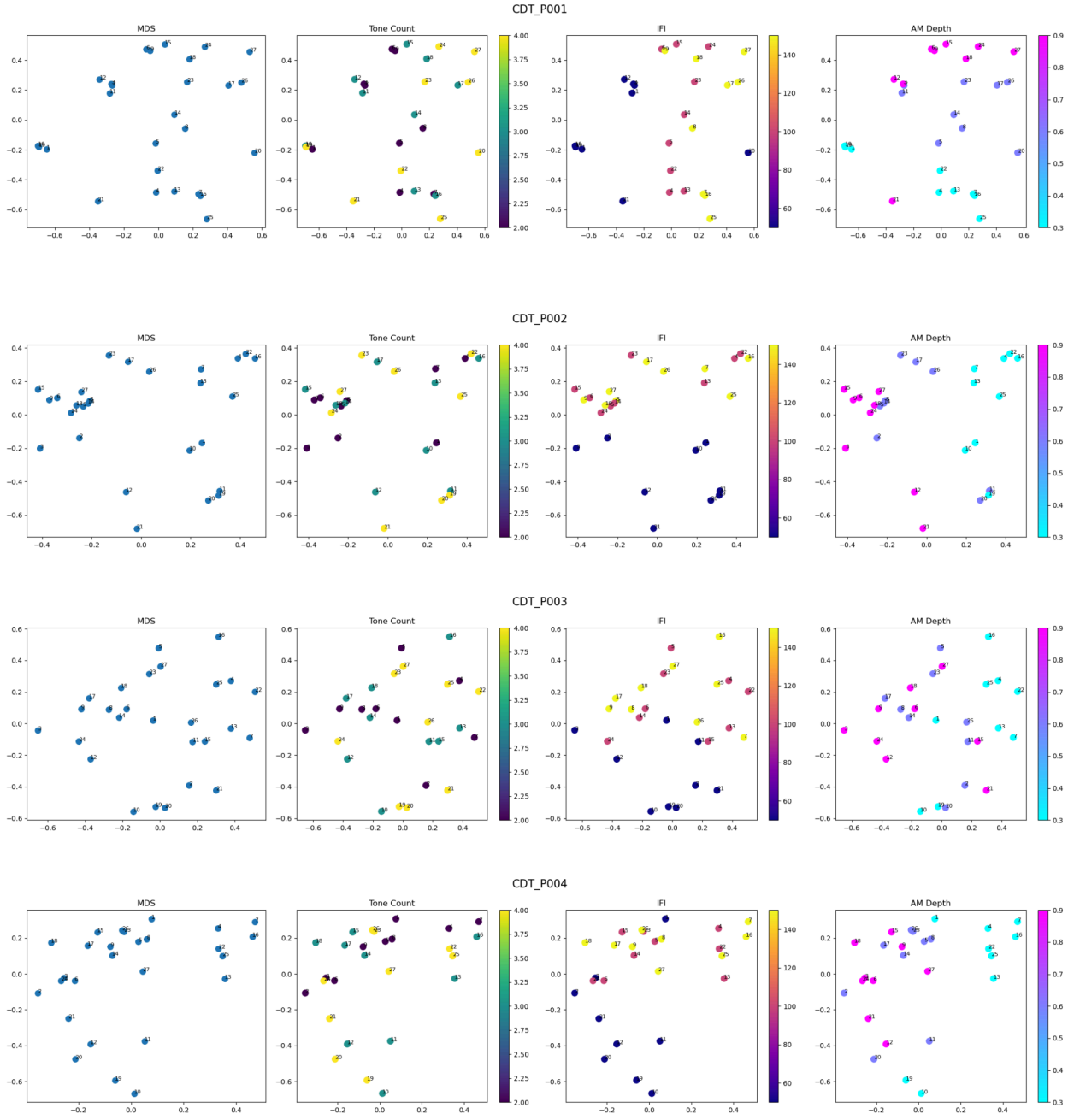


Figure 3. Group MDS configuration coloured according to (A) Inter-Frequency Interval (IFI), (B) Number of Tones, and (C) Amplitude Modulation (AM) Depth.

3.1.4 Individual MDS Solutions

Individual MDS solutions were computed for each participant to examine the consistency of the recovered perceptual organization (Figure 4). Although the precise spatial arrangement varied across participants, similar trends emerged in the colored MDS plots. In particular, AM depth produced relatively consistent clustering across participants, while IFI showed moderate organization, and Tone Count exhibited comparatively weaker separation. These findings indicate that the overall perceptual

structure recovered from the group analysis was not driven by a single participant but rather reflected patterns broadly shared across observers.



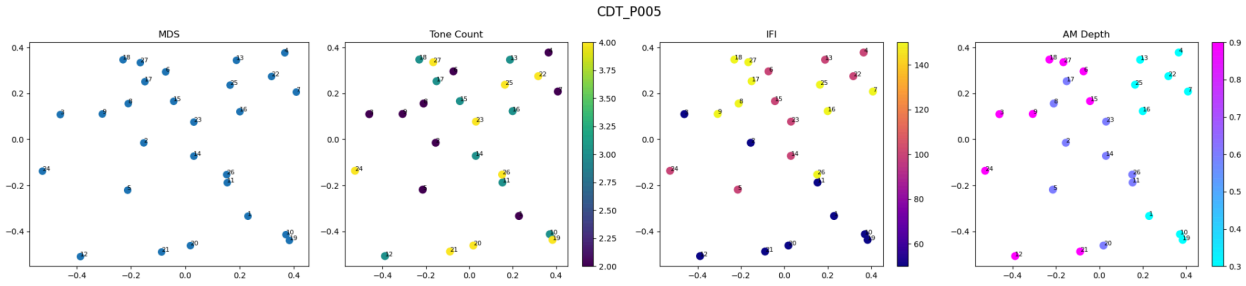


Figure 4. Individual participant MDS configurations together with the corresponding stimulus parameter overlays.

3.2 Pilot 2: Effect of Waveform, IFI, and AM Depth on Perceptual Organization

3.2.1 Selection of the MDS Solution

Stress values were computed for one- to four-dimensional MDS solutions (Figure 5). A sharp reduction in stress was observed from one to two dimensions, while the improvement beyond two dimensions was comparatively small. Therefore, a two-dimensional solution was selected for visualization and interpretation of the perceptual space.

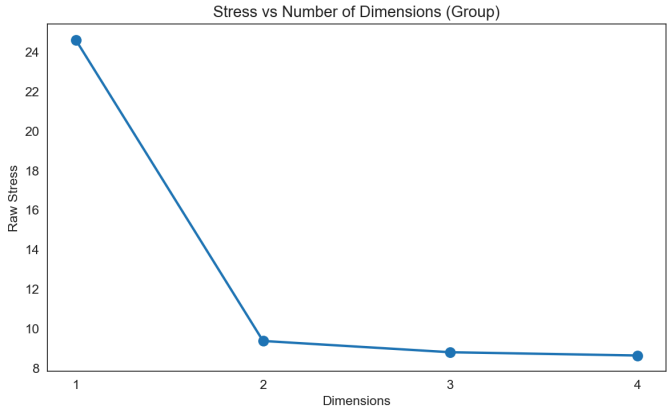


Figure 5. Stress values for one- to four-dimensional MDS solutions for Pilot 2

3.2.2 Group Perceptual Space

The two-dimensional MDS configuration is shown (Figure 6). The recovered perceptual space demonstrated a structured arrangement of the 27 stimuli, with several neighboring clusters emerging rather than a random spatial distribution.

To facilitate interpretation, the recovered perceptual configuration was coloured according to the manipulated stimulus parameters.

The waveform-coloured representation showed considerable overlap between sine, square, and sawtooth stimuli, suggesting that waveform alone did not produce strong categorical separation. In contrast, coloring by inter-frequency interval (IFI) revealed a clear gradient across the perceptual space, with stimuli of similar IFIs occupying neighboring regions. A similar trend was observed for amplitude modulation (AM) depth, with stimuli of comparable depth clustering together. These observations suggest that IFI and AM depth contributed more strongly to perceived similarity than waveform shape.

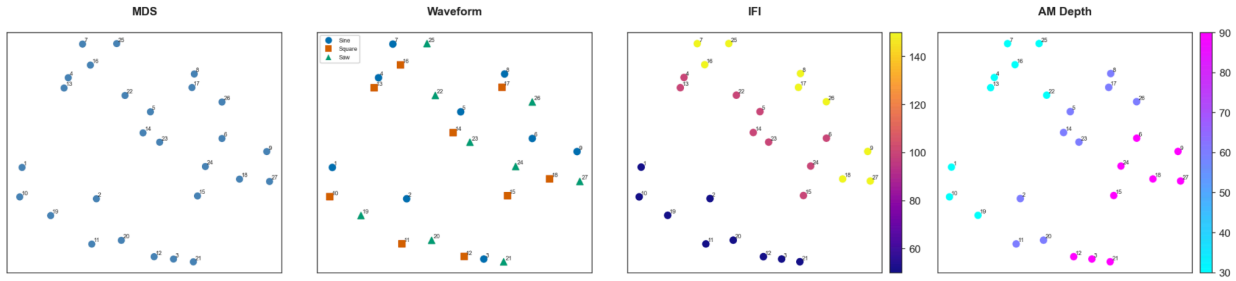
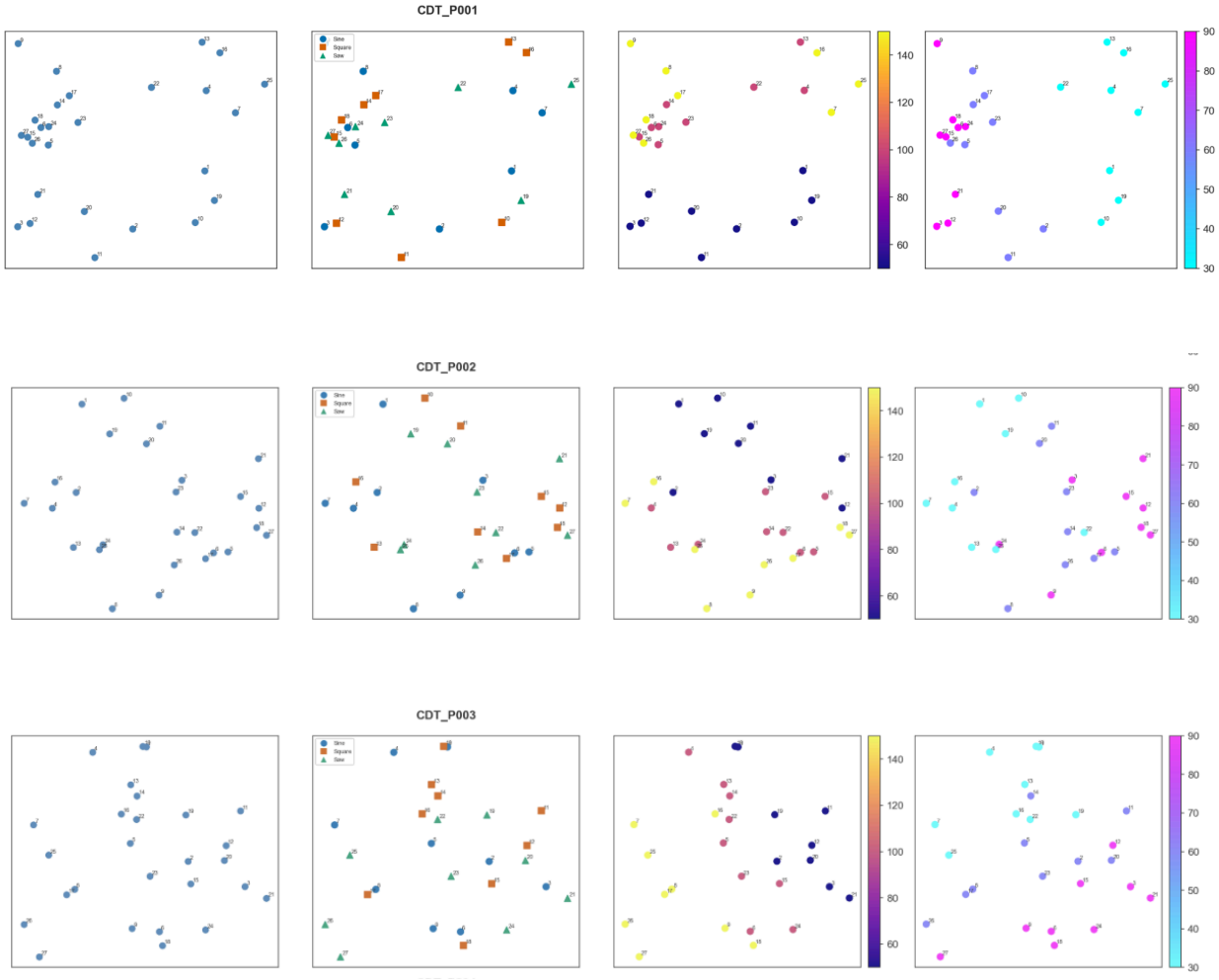


Figure 6. Two-dimensional MDS solutions for pilot 2 and group MDS configuration coloured according to (A) Waveform Shape, (B) Inter-Frequency Interval (IFI), and (C) Amplitude Modulation (AM) Depth.

3.2.3 Individual MDS Solutions

Individual MDS solutions for all six participants are shown (Figure 7). Although the precise arrangement of stimuli varied between participants, the overall organization remained broadly consistent.

When individual solutions were coloured according to waveform, substantial overlap between waveform categories was observed across participants, indicating that waveform was not consistently used as the primary basis for similarity judgments. In contrast, coloring by IFI and AM depth revealed relatively consistent gradients across participants, with neighboring stimuli frequently sharing similar parameter values. This consistency suggests that participants relied more heavily on temporal and spectral characteristics associated with IFI and AM depth than on waveform identity.



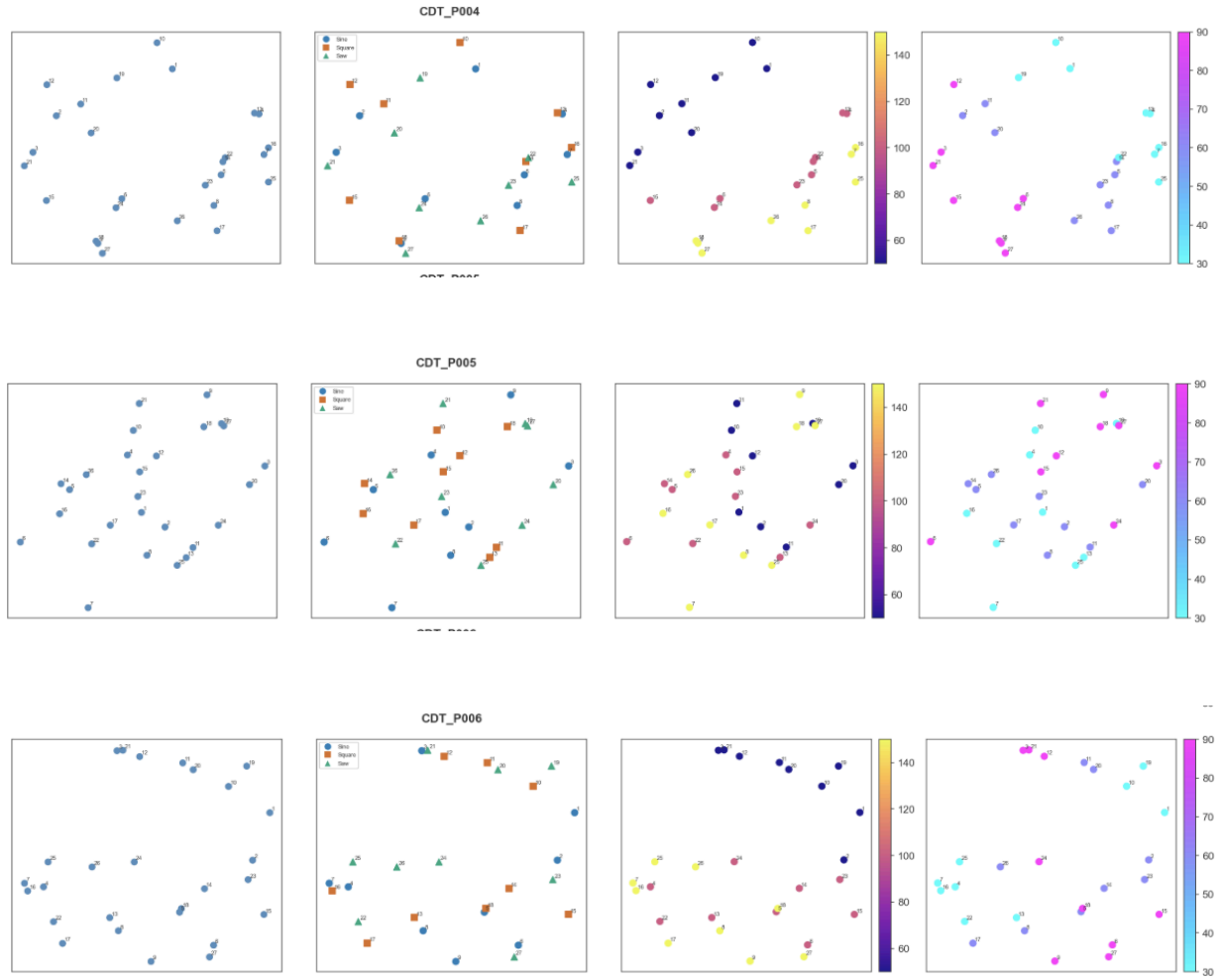


Figure 7. Individual participant MDS configurations together with the corresponding stimulus parameter overlays.

3.2.4 Verbal Descriptions of the Perceptual Space

Following the similarity judgment task, participants were asked to verbally describe the sensations associated with each stimulus. Their responses were projected onto the recovered MDS space (Figure 8) to aid interpretation of the perceptual dimensions.

Despite differences in vocabulary, several recurring perceptual themes emerged across participants. Many descriptions referred to speed or frequency, using terms such as fast, slow, continuous, and pulsing.

Others emphasized intensity or force, including descriptions such as hard, high intensity, more pressure, rough, and strong. A third set of responses focused on texture or temporal pattern, with descriptors such as smooth, break, beep, engine sound, drizzle, and heartbeat.

Interestingly, participants often used analogous descriptors despite not being instructed to adopt a common vocabulary. This convergence suggests that the recovered perceptual dimensions capture meaningful perceptual characteristics rather than arbitrary stimulus relationships. Overall, the verbal labels indicate that participants primarily organized the stimuli by perceived temporal characteristics and intensity-related qualities.

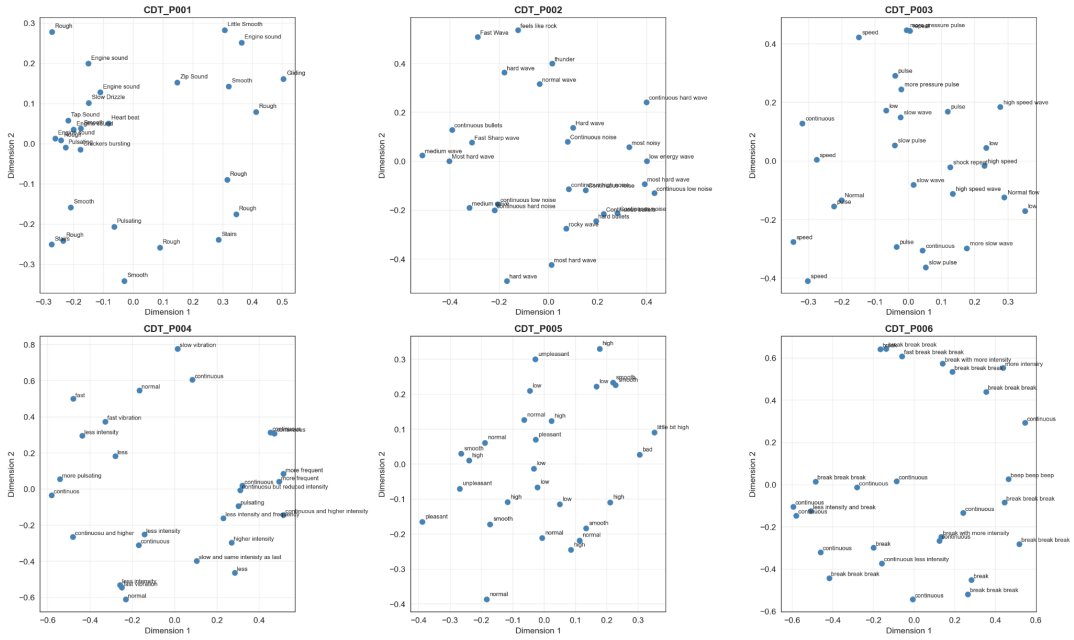


Figure 8. Verbal descriptions were projected onto the respective participant MDS configurations for Pilot 2.

3.3 Pilot 3: Effect of Attack Time, IFI, and AM Depth on Perceptual Organization

3.3.1 Stress Analysis

Stress values were computed for one- to five-dimensional MDS solutions (Figure 9). A sharp reduction in stress was observed from one to two dimensions, while the improvement beyond two dimensions was comparatively small. Therefore, a two-dimensional solution was selected for visualization and interpretation of the perceptual space.

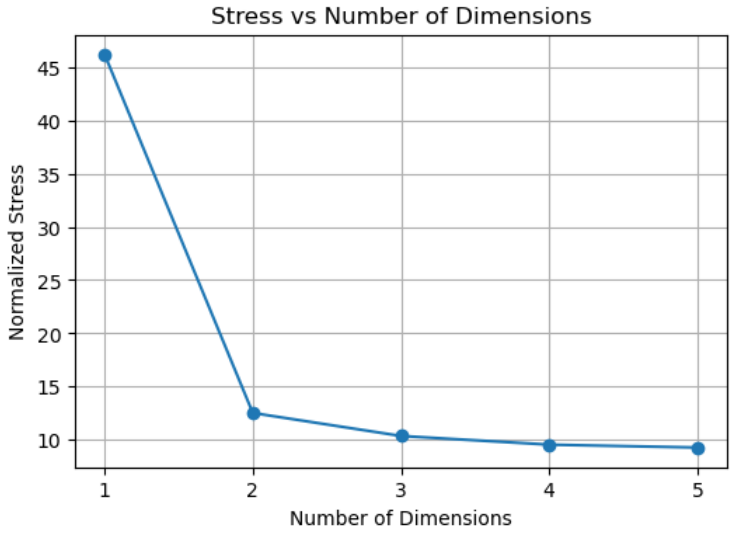


Figure 9. Stress values for one- to four-dimensional MDS solutions for Pilot 3.

3.3.2 Group Perceptual Space

The two-dimensional group MDS solution is shown (Figure 10). Unlike the previous pilot, the present stimulus set was designed using Attack Time, Inter-Frequency Interval (IFI), and Amplitude Modulation (AM) Depth as the primary stimulus parameters.

To interpret the recovered perceptual space, the MDS configuration was colored by each experimental parameter (Figure 11).

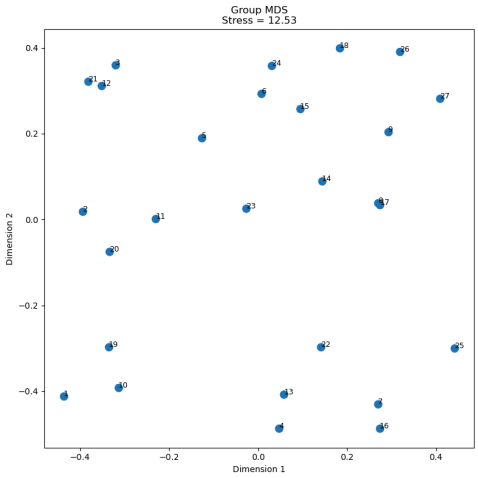


Figure 10. Two-dimensional group MDS configuration for Pilot 3.

The color gradients indicate a systematic organization of the perceptual space. Stimuli with similar IFI values occupied neighboring regions, forming a gradual gradient across the configuration. AM Depth also exhibited clear clustering, with stimuli sharing similar modulation depths tending to occupy nearby locations. In comparison, Attack Time showed weaker, less consistent organization, with substantial overlap across the different attack-time levels. These observations suggest that IFI and AM Depth had a greater influence on participants' similarity judgments than Attack Time.

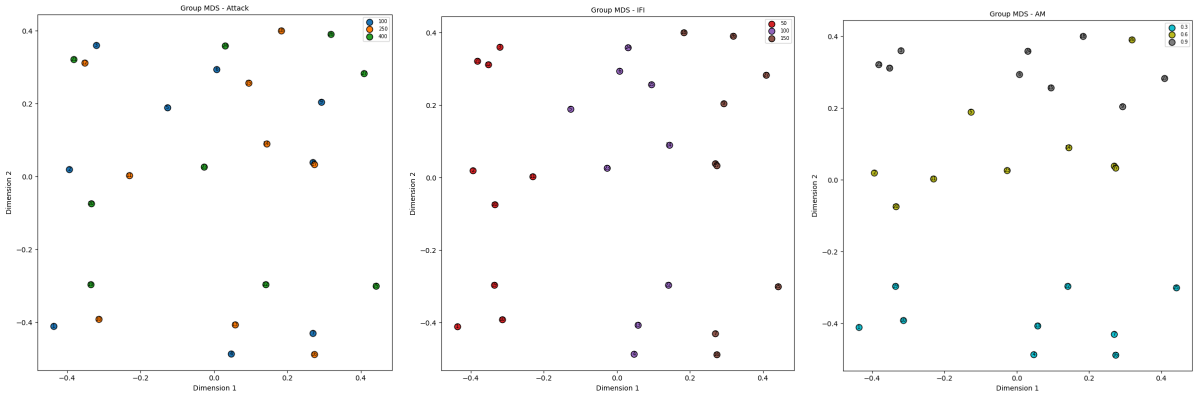


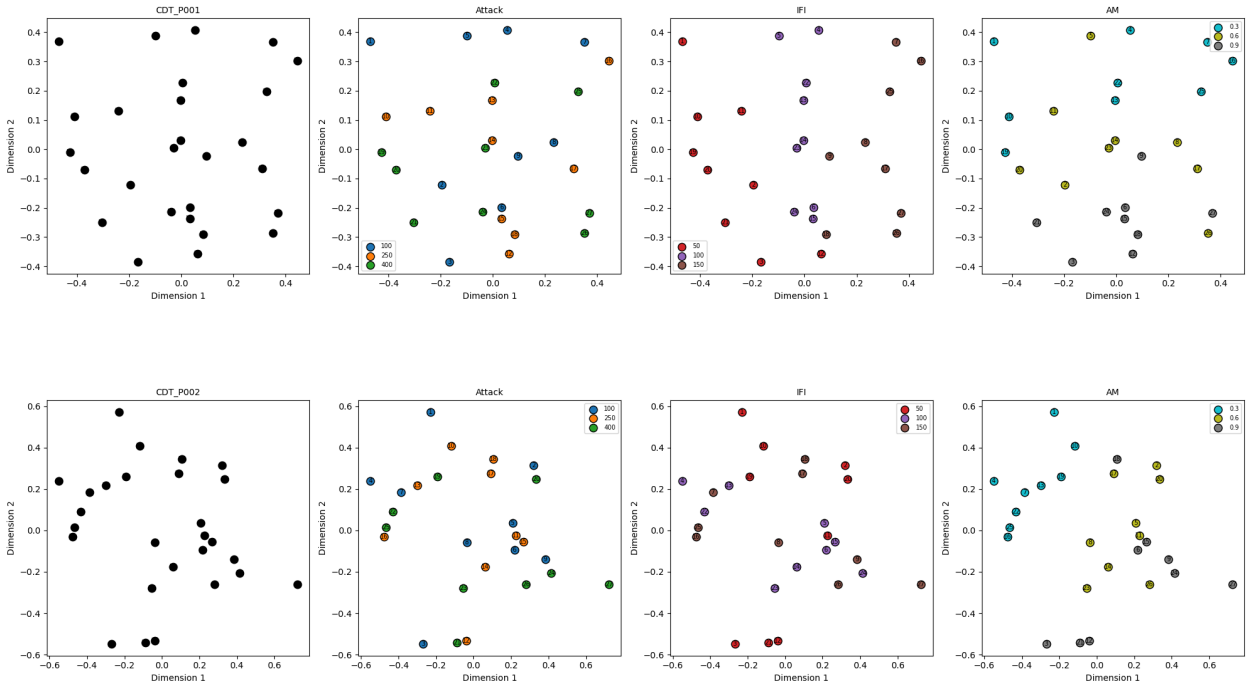
Figure 11. Group MDS configuration coloured according to (A) Attack Time, (B) Inter-Frequency Interval (IFI), and (C) Amplitude Modulation (AM) Depth.

3.3.3 Individual MDS Solutions

Individual MDS configurations for each participant are presented (Figure 12).

Although the exact spatial arrangement differed slightly across participants, similar organizational trends were observed across individual solutions. Coloring the configurations by manipulating stimulus parameters showed that neighboring stimuli frequently shared similar IFI and AM depth values. Attack time exhibited a weaker clustering pattern across several participants.

The consistency between individual and group solutions indicates that the recovered perceptual organization was not driven by a small subset of observers but reflected shared perceptual strategies across participants.



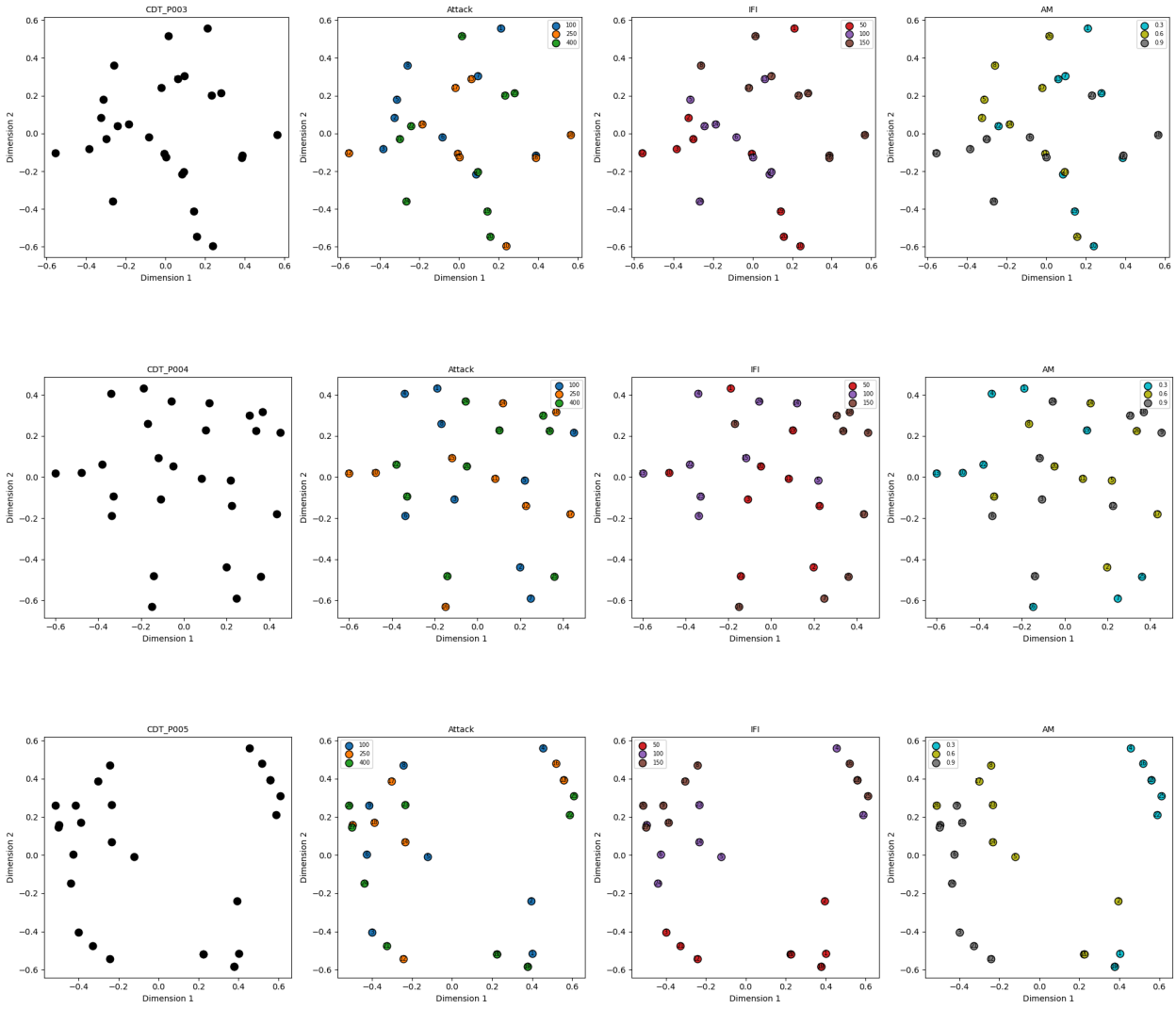


Figure 12. Individual participant MDS configurations together with the corresponding stimulus parameter overlays.

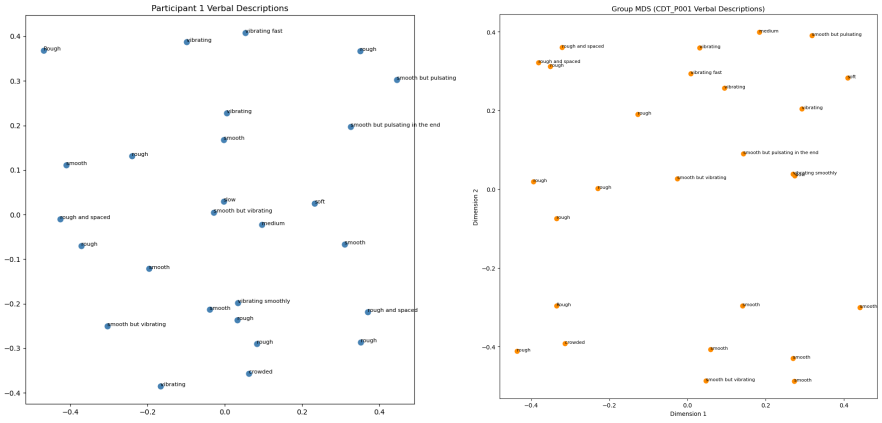
3.3.4 Verbal Descriptions

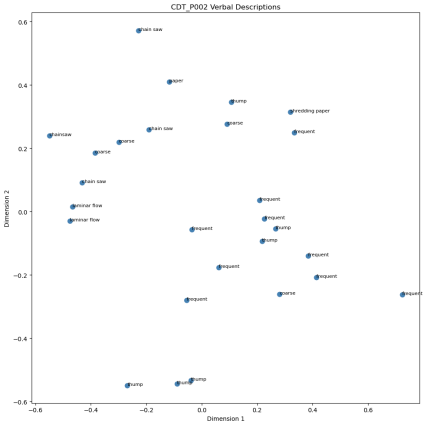
Participants were asked to provide verbal descriptions for each stimulus after completing the similarity judgment task. These descriptions were projected onto both the individual and group MDS solutions to aid interpretation of the recovered perceptual dimensions (Figure 13).

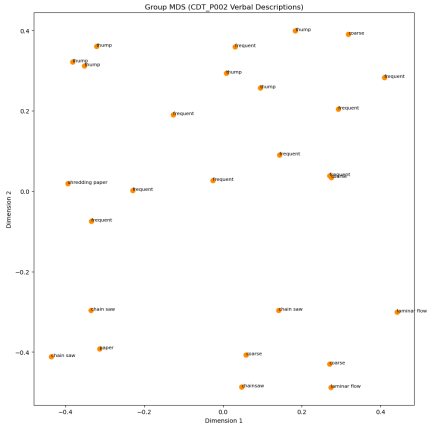
Across participants, several recurring descriptive themes emerged despite differences in wording. Common descriptors included rough, smooth, hard, soft, high-intensity, low-intensity, vibrating, frequent, coarse, bumpy, pressing, and shock-like sensations. Some participants also described stimuli using familiar real-world analogies such as chainsaw, paper, or pressing a finger, suggesting that they naturally associated the vibrotactile signals with everyday tactile experiences.

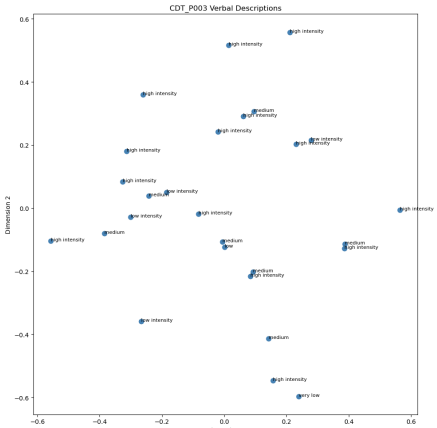
The verbal descriptions showed good agreement with the group MDS solution, with similar perceptual descriptors corresponding to nearby regions in the recovered perceptual space.

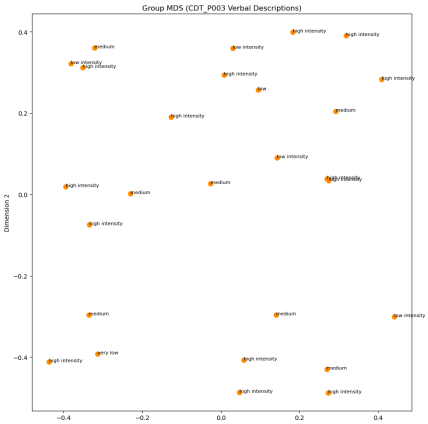
Compared to Pilot 2, the verbal descriptions in this pilot were more consistent and primarily reflected variations in perceived intensity, roughness, temporal frequency, and surface texture. These qualitative observations align with the organization observed in the MDS configuration and further support the selection of Attack Time, IFI, and AM Depth as perceptually meaningful stimulus dimensions.

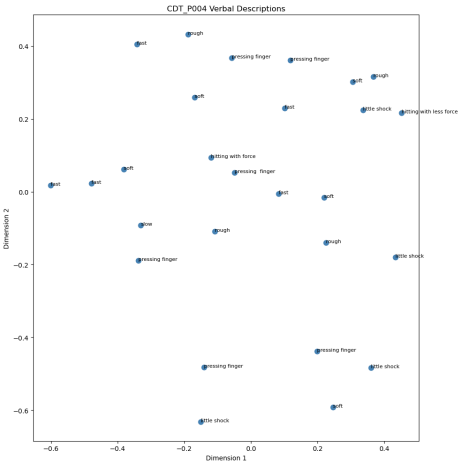


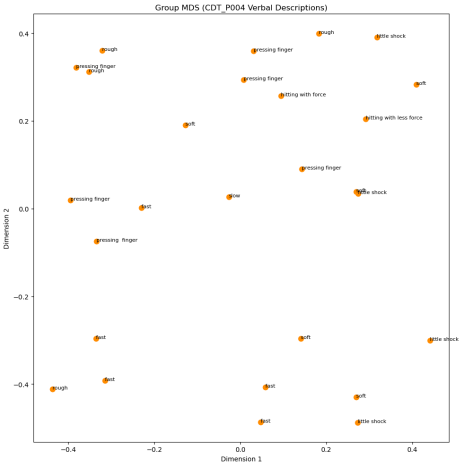

CDT_P002 Verbal Descriptions
Dimension 1
Dimension 2


Group MDS (CDT_P002 Verbal Descriptions)
Dimension 1
Dimension 2


CDT_P003 Verbal Descriptions
Dimension 1
Dimension 2


Group MDS (CDT_P003 Verbal Descriptions)
Dimension 1
Dimension 2


CDT_P004 Verbal Descriptions
Dimension 1
Dimension 2


Group MDS (CDT_P004 Verbal Descriptions)
Dimension 1
Dimension 2

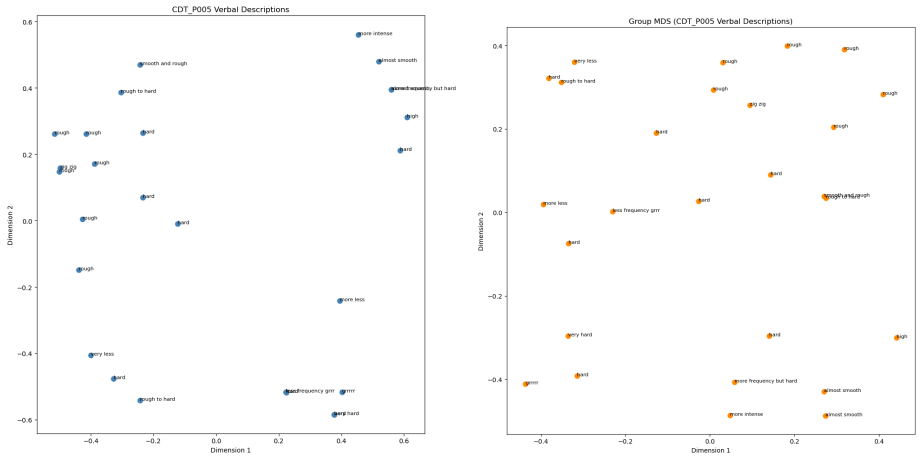


Figure 13. Verbal descriptions projected onto their respective MDS configuration (left) and group MDS configuration (right) for Pilot 3.

4. Discussion

Across all three pilot experiments, two stimulus parameters consistently contributed to the perceptual organization of complex vibrotactile stimuli: Inter-Frequency Interval (IFI) and Amplitude Modulation (AM) Depth. In contrast, the third manipulated parameter in each pilot, Number of Tones, Waveform Shape, or Attack Time, produced comparatively weaker perceptual separation within the multidimensional scaling solutions.

The consistent influence of IFI is supported by previous studies demonstrating that the tactile system is sensitive to changes in the spectral composition of complex vibrations rather than individual frequency components alone. Le et al. showed that altering the spacing of frequency components alters perceived similarity, even when overall intensity is controlled, suggesting that observers use spectral information when discriminating complex vibrotactile signals (Le et al., 2024). Because IFI alters the spacing between frequency components and consequently modifies broader spectral characteristics, including spectral distribution and, depending on the stimulus configuration, spectral centroid, and harmonic relationships, which likely contributed to the consistent clustering observed across all three pilots.

Similarly, the strong influence of Amplitude Modulation (AM) Depth agrees with previous findings that temporal envelope cues play a major role in vibrotactile perception. Chauvette et al. (2025) demonstrated that temporal features, including modulation characteristics, are salient perceptual dimensions in both auditory and vibrotactile timbre perception. The present findings suggest that participants were consistently sensitive to differences in modulation depth, resulting in clearer perceptual organization than with several other manipulated stimulus parameters.

In contrast, the number of Tones, Waveform Shape, and Attack Time produced comparatively weaker perceptual organization. Several explanations may account for these findings. First, these parameters may have produced perceptual changes that were less salient than the spectral and temporal cues provided by IFI and AM Depth. Second, interactions between multiple simultaneously varying stimulus dimensions may have reduced the perceptual contribution of individual parameters. Finally, the relatively small sample sizes used in these pilot studies may have limited the emergence of stronger clustering for these dimensions. Further experiments with larger participant groups and refined stimulus sets are required to determine whether these parameters contribute independently to vibrotactile similarity judgments.

The verbal descriptions further support these findings. Participants generally relied on holistic perceptual descriptors, including rough, smooth, sharp, soft, and vibrating. Similar observations have been reported in studies of tactile texture perception, where perceptual judgments are often organized around subjective sensory qualities rather than explicit physical stimulus characteristics (Picard et al., 2003).

Overall, the three pilot experiments suggest that the perceptual space of complex vibrotactile stimuli is influenced predominantly by spectral and temporal modulation cues. Among the parameters investigated, IFI and AM Depth consistently accounted for the greatest perceptual organization, providing useful directions for designing future vibrotactile stimuli and identifying the dimensions that most strongly shape tactile similarity judgments.

The iterative design of the three pilots also demonstrates the usefulness of MDS during stimulus development. Rather than simply evaluating discrimination performance, MDS enabled visualization of the perceptual organization of increasingly refined stimulus sets, providing a principled basis for selecting stimulus dimensions for future categorization experiments.

Limitations and Future Work

The present study was exploratory and involved a relatively small number of participants; therefore, the recovered perceptual spaces should be interpreted as preliminary. In addition, the stimuli were validated primarily through manual inspection, as an accelerometer was not available to objectively verify the actuator output. Future studies should incorporate accelerometer-based validation to ensure that the delivered vibrations accurately match the intended stimulus parameters. Furthermore, the analyses relied primarily on MDS and qualitative interpretation. Future work should include larger participant samples, additional statistical analyses to quantitatively relate stimulus parameters to perceptual similarity, and the investigation of a broader range of temporal and spectral features. Finally, the stimulus set developed in this project provides a foundation for future studies on vibrotactile categorization, perceptual learning, and the dimensional weighting of complex vibrotactile stimuli.

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